

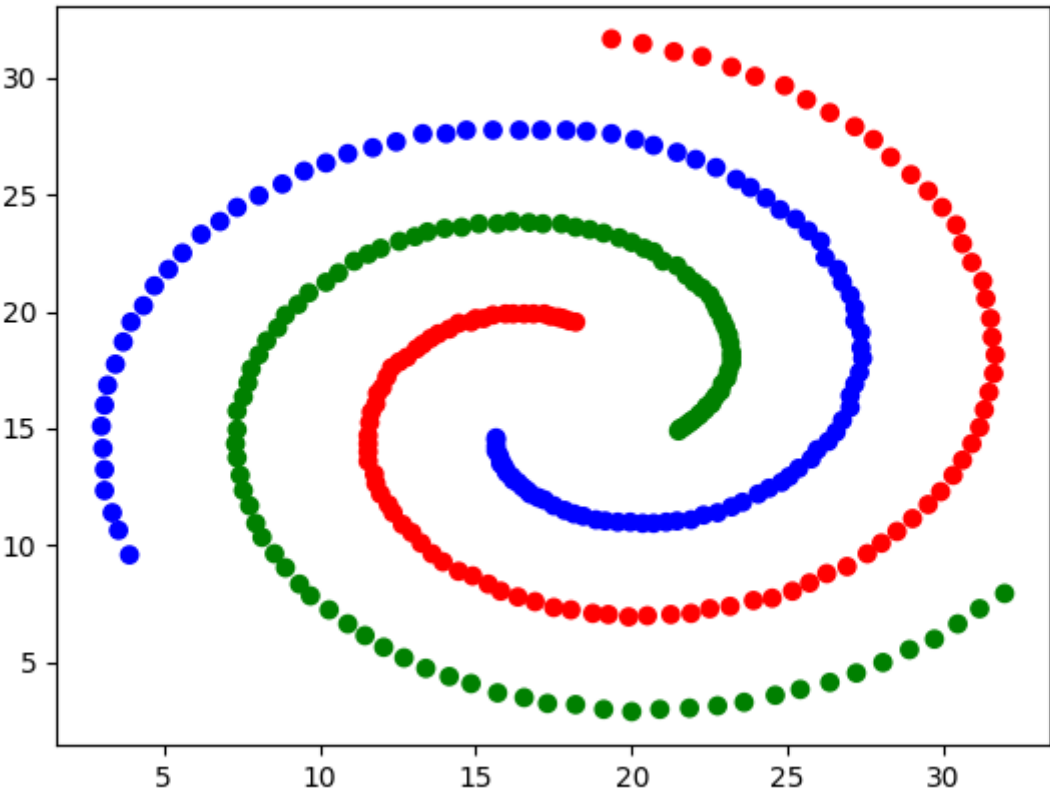
Assignment 2

Best Results for SSE and RI over 10 runs using Euclidean Distance, Cosine Similarity, and L3 Distance

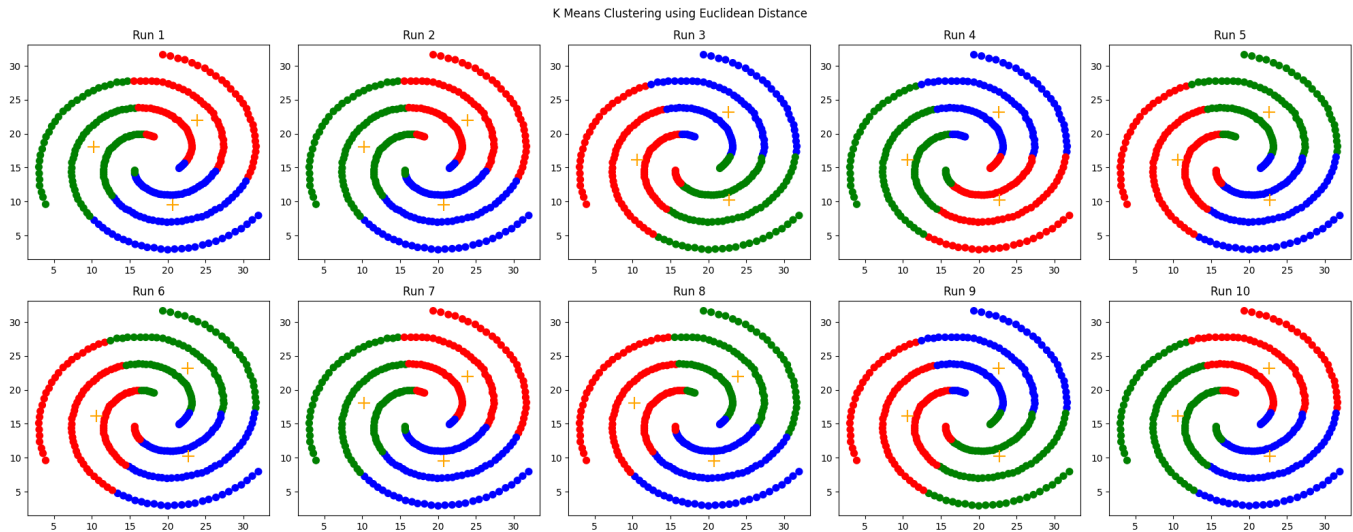
Distance Type	Best SSE	Best RI
Euclidean Distance	12313.2751	0.5544
Cosine Similarity	17045.2286	0.5624
L3 Distance	12299.1322	0.5547

The results show very similar performance for all three distance calculations for Rand Index, with Cosine Similarity performing slightly better than L2 and L3 distance. However, Cosine Similarity performs far worse than Euclidean Distance and L3 Distance for Sum of Squared Errors. This may be due the fact the SSE uses euclidean distance when comparing a point to its centroid, so this metrics may have bias towards more typical distance calculations.

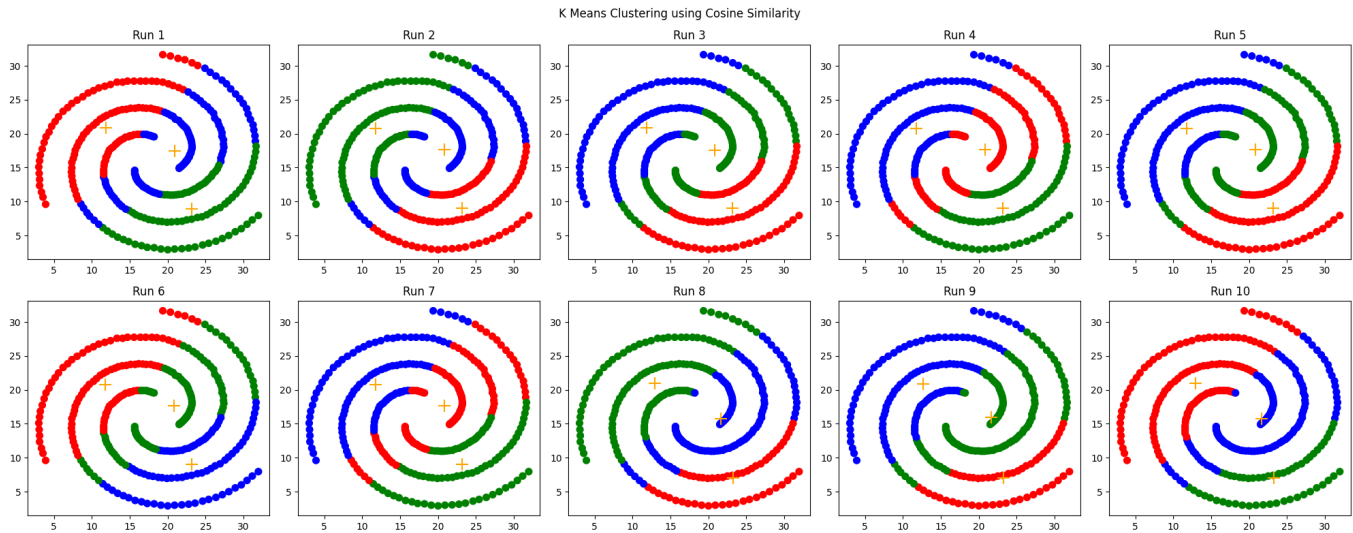
Task 1



Euclidean Distance



Cosine Similarity



L3 Distance

